Outputs Explanation

The document goes through every output file produced by the programs in the source code archive and explains what is inside it, how to read it, and what it told us. Figures are described but not shown here; the files themselves sit in the numbered output folders.

q2_age_outlier_sensitivity.py

The PCA flagged three wild-type samples that sat apart from their group. The worry was that the whole Q2 ageing result might be driven by those three fish rather than by ageing. The script re-runs the Q2 comparison with them removed and compares the two versions.

- **q2_outlier_sensitivity_comparison.csv:** one row per gene with its log2 fold change and q-value from the full run and from the reduced run side by side, plus flags for whether the gene was significant in each run and whether the two runs agree on the direction of change. This is the table that lets me say the result is or is not robust, rather than just hoping it is.
- **q2_age_DEG_results_no_outliers.csv:** the complete DE results for the reduced run, in the same format as the original Q2 table, so the two can be compared directly or the reduced version can be used on its own.
- **q2_outlier_sensitivity.png:** two panels. The first is a scatter of the fold changes from the full run against the fold changes from the reduced run, where points lying on the diagonal mean the two runs agree. The second is a bar chart of how many DEGs each run produced. The points sitting on the diagonal and all 515 original DEGs maintaining their direction confirms the outliers were not driving the result. Notably, removing the three samples raises the DEG count to 1,219 rather than reducing it, because those animals carry a large share of the within-group variance. Retaining them is therefore the conservative choice, and 515 is the primary reported figure.